Integrating Spatial Context With Generative Model for Building Reconstruction From Airborne LiDAR Point Clouds

Yuxiang Zhao, Shengwen Li, *Member, IEEE*, Fang Fang, *Member, IEEE*, Nan Min, Sishi Gong, Yu Wang, and Shunping Zhou, *Member, IEEE*

***Abstract*—Reconstructing 3-D buildings from airborne light detection and ranging (LiDAR) point clouds (PCs) is a fundamental task with critical importance in applications such as urban planning, autonomous driving, and emergency services. The emerging methods based on the generative model have achieved remarkable progress in this task. However, previous methods fail to fully exploit the spatial context inherent in PCs, leading to inaccuracies in building reconstruction. This study proposes a novel method named SCGNet that integrates spatial context with the generative model to improve the reconstruction of 3-D building models. Specifically, a global context-based vertex generation (GCVG) module is first introduced to predict building vertices. Then, a vertex-centric multiscale spatial context encoding (VMSCE) module is developed to capture the multiscale semantics of spatial context that is centered on building vertices. Finally, a local context-enhanced face generation (LCFG) module is designed to reconstruct building faces. Extensive experiments demonstrate that the proposed method outperforms the baselines and is robust. This study explores a novel approach for optimizing building reconstruction and provides a methodological reference for PC-based 3-D model tasks. The source code, model weights, and dataset will be available at: https://github.com/ShuaibZyx/SCGNet.**

***Index Terms*—Airborne light detection and ranging (LiDAR), building reconstruction, generative model, point clouds (PCs), spatial context.**

I. Introduction

THE 3-D building reconstruction plays a vital role in numerous applications, including urban planning, autonomous driving, emergency response, and virtual reality [1], [2], [3], [4]. Recently, continuously accumulating airborne light detection and ranging (LiDAR) point clouds (PCs) has emerged as a critical data source for the reconstruction of 3-D buildings [5], [6], [7], [8], [9]. These PCs exhibit great advantages in capturing complex building geometries and demonstrate considerable potential for accurate 3-D building reconstruction [10], [11], [12], [13]. In practice, building structures in PCs exhibit substantial complexity, characterized by irregular geometries, frequent occlusions, and nonuniform point densities. Accurately reconstructing 3-D building models from LiDAR PCs is still challenging.

Nowadays, segmentation-based methods are widely used to automatically reconstruct 3-D building models from airborne LiDAR PCs [9], [14], [15]. For example, RoofNet [16] employed the PointNet++ [17] backbone to develop a neural network that simultaneously segmented individual roof planes and clustered them with similar geometric shapes, demonstrating promising performance. And, geometric feature points such as corners and boundaries have been used to establish relationships between points and roof plane segments, facilitating accurate building reconstruction [18]. In addition, a boundary-aware point clustering approach has been introduced to segment roof planes in multiple embedding spaces [19]. However, segmentation-based methods still face limitations in reconstructing buildings accurately since the LiDAR PCs of buildings are incomplete and contain outliers, especially when dealing with complex roofs with compact structures.

Recently, generative model-based methods have gained attention as a promising approach to address the aforementioned limitations [20], [21]. In particular, Point2Building (P2B) [22] represented a learning-based method that directly predicted 3-D building models from airborne LiDAR PCs via an autoregressive model. P2B improved adaptability to various building geometries by iteratively generating sequences of mesh vertices and faces. However, it does not fully leverage the spatial context inherent in PCs, which leads to inaccuracies in building reconstruction.

This article proposes a generative method named SCGNet that integrates the spatial context of PCs to improve the 3-D building reconstruction from LiDAR PCs. Specifically, the proposed method first introduces a global context-based vertex generation (GCVG) module to predict the building vertices. And, a vertex-centric multiscale spatial context encoding (VMSCE) module is developed to capture the multiscale semantics of spatial context that is centered on building vertices. Finally, a local context-enhanced face generation (LCFG) module is presented to reconstruct building faces. To the best of authors' knowledge, this study

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is the first attempt that integrates both the global–local context of LiDAR PCs and the generative model for building reconstruction.

The main contributions of this study are summarized as follows.

1) This study highlights the spatial context inherent in PCs and proposes to integrate global–local context with the generative model to improve 3-D building reconstruction.
2) A novel method named SCGNet is proposed, which encodes spatial context centered on building vertices for facilitating building face generation.
3) Extensive experiments demonstrate that the proposed method outperforms existing baselines and is robust.

II. Related Work

A. Building Reconstruction via Primitive Fitting

The early methods reconstruct 3-D building models by fitting building structures using predefined geometric primitives from PCs [23]. For example, Verma et al. [24] introduced a roof topology graph to represent the relationships among different roof primitives and used subgraph recognition to identify the primitives that compose the roof structure. Henn et al. [25] employed a support vector machine (SVM) to identify roof primitives and reconstruct building models from sparse PCs. Xiong et al. [26] analyzed the topological properties of building surfaces to search for suitable predefined primitives. Chen et al. [27] reconstructed highly regular and semantically rich building models through primitive clustering. Zhang et al. [28] proposed an optimization method that reconstructed complex roofs by minimizing the distance between PCs and predefined primitives. Although these methods simplify reconstruction through architectural priors, they are insufficient to exhaustively enumerate various geometric elements, which limits their adaptability to diverse structures, especially buildings with irregular components.

Recently, some methods have employed advanced algorithms to extract diverse geometric primitives from PCs, thereby improving the fitting accuracy of building models [29], [30]. In practice, algorithms such as random sample consensus (RANSAC) and region-growing are widely employed to extract diverse geometric primitives. The RANSAC-based methods randomly sampled subsets of points to extract multiple candidate planes and selected the best fit as the roof component. For example, Xu et al. [31] proposed an enhanced RANSAC algorithm incorporating a weighting mechanism for roof plane extraction. Li et al. [32] proposed an improved RANSAC algorithm based on normal distribution transformation (NDT) cells, using NDT cells as the minimal iterative unit to enhance sampling quality. Li and Shan [33] proposed the multiprimitive reconstruction (MPR), which employed a two-step RANSAC algorithm to identify planar primitives within complex building boundaries.

Alternatively, the methods based on region-growing started from seed points selected based on geometric criteria and iteratively expanded to neighboring points to form planar segments. For instance, Yan et al. [34] detected roof planes using region-growing from PCs and refined the results through global energy optimization. Wang et al. [35] reconstructed watertight building models from candidate planes extracted by region-growing with the structural constraints. Li et al. [36] improved the region-growing algorithm by incorporating hierarchical clustering and boundary relabeling to extract roof planes from airborne LiDAR PCs. However, the geometric primitives extracted by such algorithms still often struggle to capture detailed geometric structures.

B. Building Reconstruction via Deep Learning

In recent years, deep learning-based discriminative methods have advanced building reconstruction by directly learning to extract fundamental geometric elements from LiDAR PCs, producing more detailed structures than traditional primitive-fitting methods. Existing methods typically predict building corners and edges from LiDAR PCs by leveraging global–local context modeling encoded in learned pointwise features. For example, PC2WF [37] formulates roof reconstruction as a joint task of corner detection and edge prediction, constructing 3-D models from surface PCs. Point2Roof [12] identifies candidate building corners using pointwise features with offset regression, followed by edge prediction on a fully connected graph. EDGE [38] proposes an edge-guided reconstruction framework that leverages local context between points and edges to improve both corner detection and edge prediction. BWFormer [39] mitigates the inefficiency of 3-D corner search in LiDAR PCs by introducing a 2-D-to-3-D corner detection strategy, combined with edge attention to refine edge prediction. In addition, RR-Net [40] employs an edge-point segmentation to extract roof structures and reconstructs 3-D models by searching closed loops in graphs formed by detected building corners. However, discriminative methods are constrained by structural occlusions and sparse PCs in LiDAR data, resulting in incomplete reconstruction.

Recent studies have employed conditional generative models to infer the overall structure of building models [11], [22], [41]. In these studies, the LiDAR PCs were encoded as global context to provide precise 3-D geometric conditions, and achieved constrained building reconstruction by iteratively generating fundamental geometric elements, such as vertices, edges, and faces. However, the generative models do not fully leverage the spatial context inherent in PCs, which leads to inaccuracies in building reconstruction. Spatial context, particularly global–local context, has shown remarkable promise in a series of discriminative methods for building reconstruction. In light of this, this study will integrate both the global–local context of LiDAR PCs and the generative model for building reconstruction.

III. Methodology

The proposed SCGNet integrates spatial context from PCs with the generative model to improve 3-D building reconstruction, as illustrated in Fig. 1, which consists of three key modules, including GCVG, VMSCE, and LCFG. The GCVG module first generates building vertices based on the global context extracted from PCs. Subsequently, the local context

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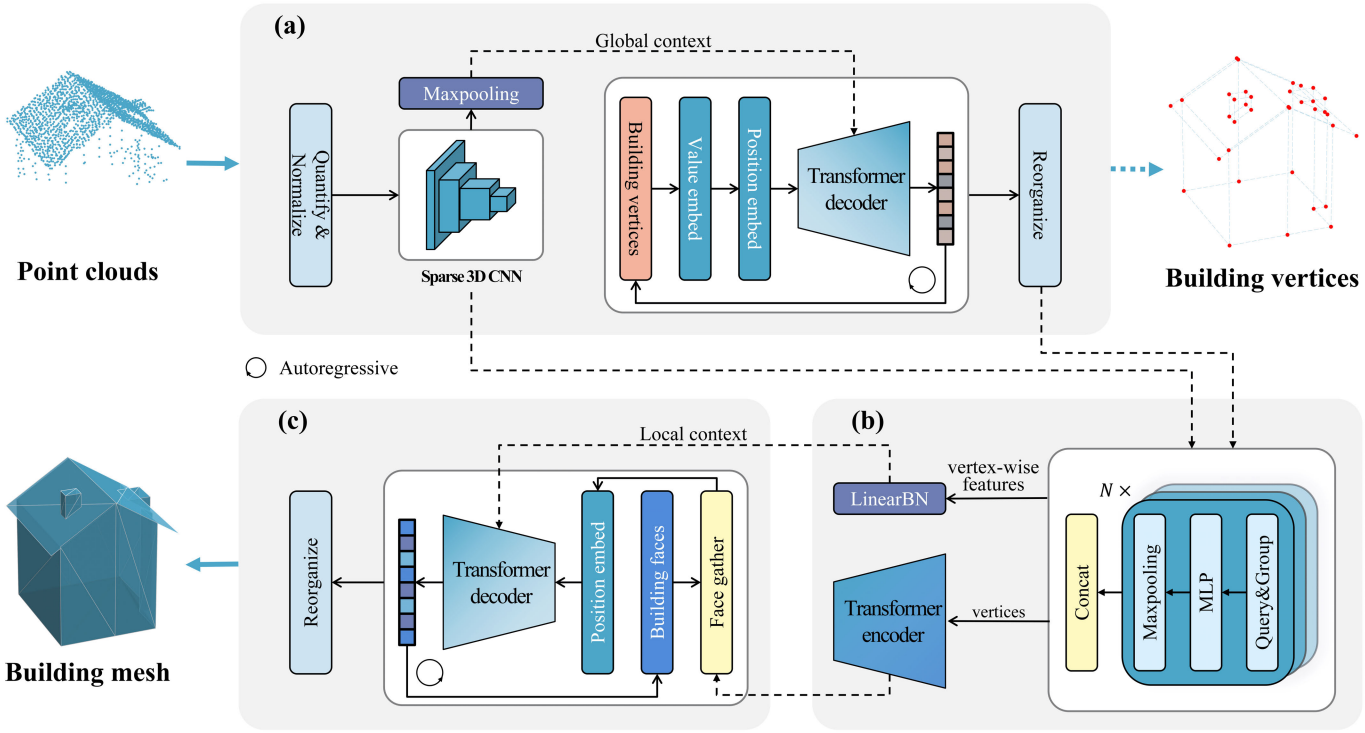


Fig. 1. Proposed method consists of three key modules: (a) GCVG; (b) VMSCE; and (c) LCFG.

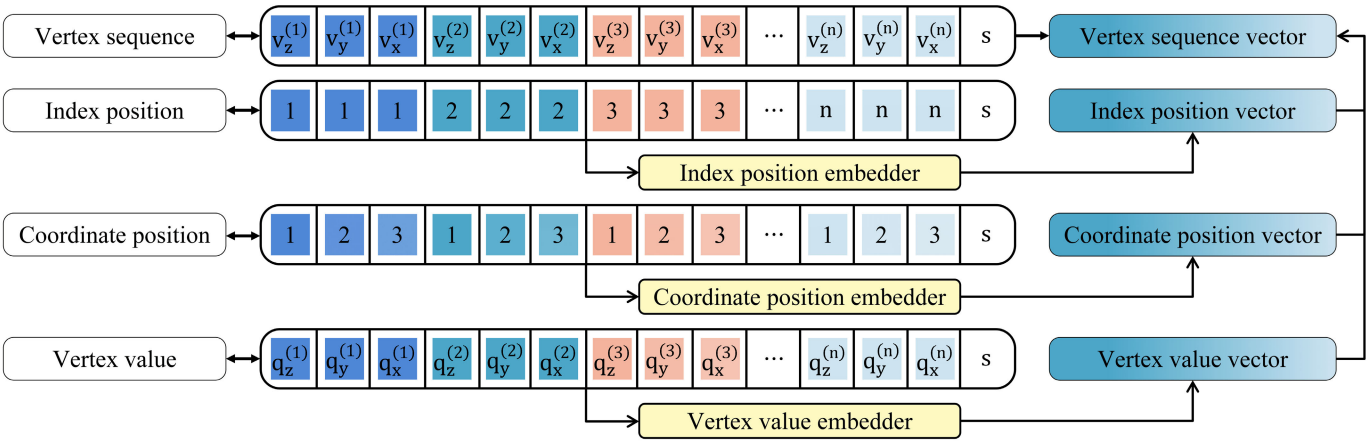


Fig. 2. Process of building vertex sequence encoding, where the building vertices are first serialized into a sequence in the order of z, y, and x, with the stop token s. Each building vertex sequence vector is encoded by three components: 1) a vertex value embedder that encodes coordinate values; 2) a coordinate position embedder that encodes vertex positions along the coordinate axes; and 3) an index position embedder that encodes the sequence position to distinguish each vertex within the sequence.

centered on building vertices is captured via the VMSCE module. Finally, the LCFG module generates building faces based on contextual guidance extracted from the VMSCE module. Sections III-A–III-D provide detailed descriptions of the three modules.

A. GCVG Module

The GCVG module takes PCs as input to generate building vertices, rather than selecting points from the PCs as building vertices. Specifically, it adopts an autoregressive generation process, where each building vertex is iteratively generated based on the previously generated vertices. The generated building vertices are quantized into discrete values to reduce floating face artifacts [21]. Here, each coordinate value $v \in \mathbb{R}^3$ is mapped to a discrete value $q \in [0, 2^n - 1]$ calculated as

$$q = \left\lfloor \frac{(v - \text{Min}) \cdot (2^n - 1)}{\text{Max} - \text{Min}} \right\rfloor = \text{Q}(v) \tag{1}$$

where $\text{Q}(\cdot)$ maps the continuous values to the discrete values of 2^n, and Min and Max denote the lower and upper bounds of the predefined coordinate range, respectively. The coordinate tuple (x, y, z) of each quantized building vertex is reordered as (z, y, x), following the design in PolyGen [42], in which z denotes the vertical axis value. Subsequently, the quantized building vertices are sorted in a predefined order to form a

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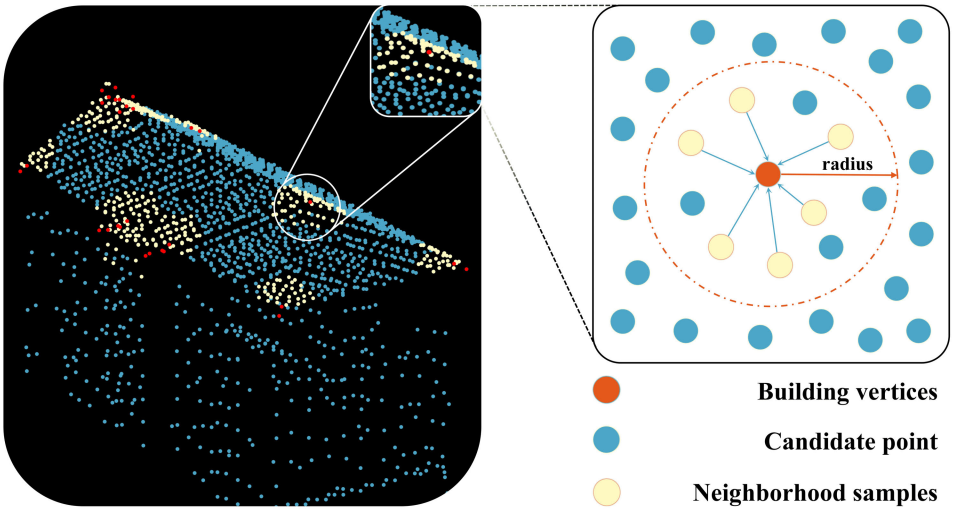


Fig. 3. Illustration of vertex-centric ball query of PCs, where the red points present the building vertices selected as centers, the blue points present candidate points, and the yellow points present the neighboring samples.

flattened sequence: from bottom to top along the z-axis, from left to right along the y-axis, and from far to near along the x-axis. After sorting, the coordinate values are flattened into a 1-D sequence as

$$\mathcal{V} = \left\{q_z^{(1)}, q_y^{(1)}, q_x^{(1)}, q_z^{(2)}, q_y^{(2)}, q_x^{(2)}, \ldots, q_z^{(n)}, q_y^{(n)}, q_x^{(n)}, s\right\} \tag{2}$$

where $q^{(i)} \in [0, 2^n - 1]$ denotes the quantized coordinate value of vertex i, and the stop token s indicates the end of the sequence.

The building vertex sequence is represented as the sum of three vectors that encode vertex coordinate values, axis-level position, and sequence-level position, as illustrated in Fig. 2. Specifically, the vertex value vector ve^{val} stores the coordinate values, the coordinate position vector ve^{coord} represents the coordinate axis corresponding to each element of the sequence, and the index position vector ve^{ind} records the position of the associated vertex within the vertex sequence. The building vertex sequence vector $e^{(v)}$ is calculated as

$$e^{(v)} = ve^{val} + ve^{coord} + ve^{ind}. \tag{3}$$

Building vertices generation is conditioned on the global context $\mathcal{Z}^{(v)}$, which is encoded from PCs using sparse 3-D convolutions [43], [44], [45]. Specifically, the PCs $\mathcal{P}^{(q)}$ of buildings are encoded into features by the network backbone B(·), while a 3-D position encoder PE(·) is used to integrate positional information. Formally, the global context $\mathcal{Z}^{(v)}$ is calculated as follows:

$$\mathcal{Z}^{(v)} = \mathrm{B}\left(\mathcal{P}^{(q)}\right) + \mathrm{PE}\left(\mathcal{P}^{(q)}\right). \tag{4}$$

The GCVG module uses a decoder to generate building vertices, which consists of 18 layers, each with four attention heads. The decoder uses building vertex sequence vectors as queries in its cross-attention calculations, while the global context $\mathcal{Z}^{(v)}$ serves as both keys and values. Then, the decoder calculates a joint probability distribution on the building vertex sequence vector $\mathcal{V}$, which is used to predict the quantization classes of the quantized coordinate values. Formally, this distribution is defined as

$$p\left(\mathcal{V} \mid \mathcal{Z}^{(v)}; \theta\right) = \prod_{n=1}^{N_v} p\left(v_n \mid v_{<n}, \mathcal{Z}^{(v)}; \theta\right) \tag{5}$$

where N_v denotes the number of building vertices that include a stop token and θ represents the module parameters. The GCVG module generates each building vertex v_n conditioned on previously generated vertices $v_{<n}$ with the global context $\mathcal{Z}^{(v)}$.

B. VMSCE Module

The VMSCE module is developed to extract vertex-centric local context, which facilitates the accurate generation of building faces. Specifically, the module first employs normalization to align the building vertices $\mathcal{V} \in \mathbb{R}^{N\times 3}$ and PCs $\mathcal{P} \in \mathbb{R}^{M\times 3}$ within the same spatial range. As illustrated in Fig. 3, a vertex-centric ball query operation is used with a radius r_k at scale k to extract the neighborhood of a building vertex. In this way, a neighborhood index $I_k \in \mathbb{N}^{N\times K}$ is constructed, where each row corresponds to the indices of the K neighborhood points. The local features $\mathcal{Z}_k^{(s)} \in \mathbb{R}^{N\times K\times(3+C)}$ are assembled based on the index I_k by gathering pointwise features $\mathcal{Z}^{(p)} \in \mathbb{R}^{M\times C}$ and their corresponding coordinates $\mathcal{P} \in \mathbb{R}^{M\times 3}$. Then, a shared multilayer perceptron (MLP) ϕ_k with max-pooling operation is used across the neighborhood points to generate a single-layer local context $\mathcal{Z}_k^{(f)} \in \mathbb{R}^{N\times C}$. Finally, the vertex-centric local context $\mathcal{Z}^{(f)}$ is calculated as

$$\mathcal{Z}_{k_i}^{(f)} = \phi_{k_i}\left(\mathcal{Z}_{k_i}^{(l)}\right) \tag{6}$$

$$\mathcal{Z}^{(f)} = \mathrm{Concat}\left(\mathcal{Z}_{k_1}^{(f)}, \mathcal{Z}_{k_2}^{(f)}, \ldots, \mathcal{Z}_{k_n}^{(f)}\right) \tag{7}$$

where n represents the hyperparameter that determines the depth of the sampling layer, and i denotes the specific scale k with the number of sampled points K_i. The vertex-centric local context $\mathcal{Z}^{(f)}$ captures the multiscale semantics of spatial context centered on building vertices.

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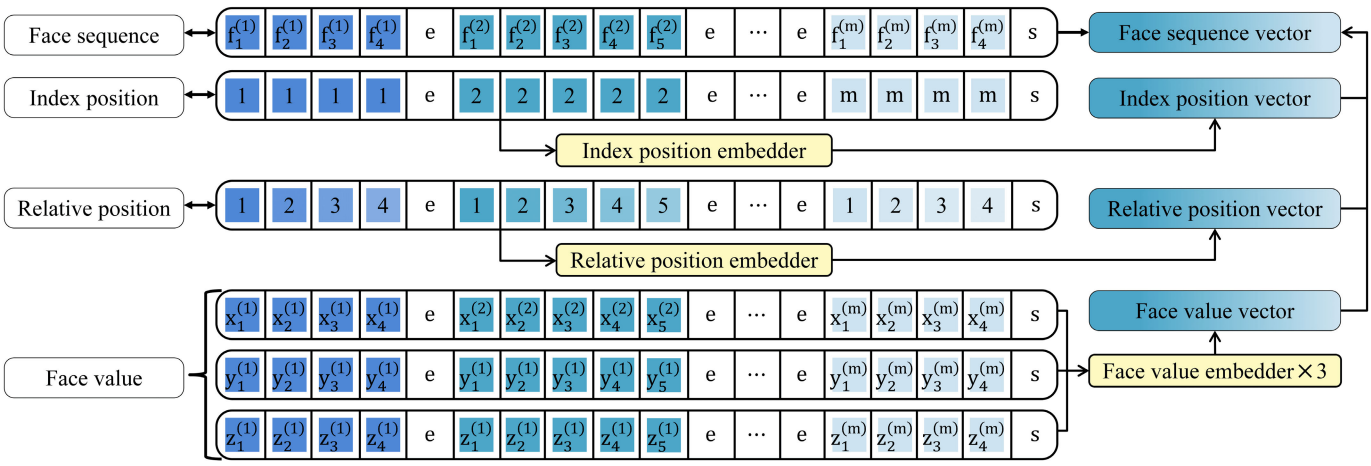


Fig. 4. Process of building face sequence encoding, where building faces are serialized into a sequence, the end token e separates individual faces, and the stop token s indicates termination. Each building face sequence vector is encoded by three components: 1) a face value embedder that encodes vertex coordinate values; 2) a relative position embedder that encodes the position of each element within a face; and 3) an index position embedder that encodes the position of each face within the sequence.

C. LCFG Module

The LCFG module uses the vertex-centric local context extracted in the VMSCE module and the generated building vertices in the GCVG module to predict building faces. The prediction of building faces is an autoregressive generation process, where each building face is iteratively predicted based on the previously generated faces and the vertex-centric local context. The indices of the generated building faces are sorted in ascending order according to the smallest vertex index, with the ties broken by the second smallest to form a flattened sequence. Then, the sorted faces, along with two special tokens, are flattened into a 1-D sequence as follows:

$$\mathcal{F} = \left\{ f_1^{(1)}, f_2^{(1)}, f_3^{(1)}, e, \ldots, e, f_1^{(m)}, f_2^{(m)}, f_3^{(m)}, f_4^{(m)}, s \right\} \tag{8}$$

where $f_i^{(j)}$ denotes the vertex index i of the face j, with e used to separate individual faces, and s indicates the end of the sequence.

The building face sequence is represented as the sum of three vectors that encode the corresponding vertex coordinate values, face-level position, and sequence-level position, as illustrated in Fig. 4. Specifically, the face value vector fe^{val} stores the vertex coordinate values corresponding to the vertex indices of each building face encoded by a Transformer encoder [46], which is composed of 12 layers, each containing four attention heads. The encoding process of the face value vector fe^{val} can be expressed as

$$\mathrm{Q}^e = \mathrm{Q}_x^e + \mathrm{Q}_y^e + \mathrm{Q}_z^e \tag{9}$$

where the vertex coordinate values Q^e, including the learnable stop tokens S, are encoded to generate the face value vector fe^{val} as follows:

$$fe^{val} = \mathrm{SelfAtt}\left(\mathrm{Concat}\left(\mathrm{S}, \mathrm{Q}^e\right)\right). \tag{10}$$

The stop tokens S allow the model to learn when to terminate the face sequence generation process. In addition, the relative position vector fe^{rel} indicates the position of each element within its corresponding face, while the index position vector fe^{ind} identifies the position of each face within the sequence. The building face sequence vector e^f is calculated as

$$e^{(f)} = fe^{val} + fe^{ind} + fe^{rel}. \tag{11}$$

Building face generation adopts a decoder based on Pointer Networks [47] to produce polygonal faces with a variable number of vertex indices. This decoder consists of 12 layers, each containing four attention heads. The decoder generates building faces represented by vertex indices through cross-attention calculations between building face sequence vectors and the vertex-centric local context $\mathcal{Z}^{(f)}$. Then, a joint probability distribution is calculated over the building face sequence vector $\mathcal{F}$, which is used to predict vertex indices in a pointer-like manner to generate each building face. This distribution is defined as

$$p\left(\mathcal{F} \mid \mathcal{V}, \mathcal{Z}^{(f)}; \theta\right) = \prod_{n=1}^{N_f} p\left(f_n \mid f_{<n}, \mathcal{V}, \mathcal{Z}^{(f)}; \theta\right) \tag{12}$$

where N_f denotes the number of vertex indices that include end and stop tokens, and θ represents the module parameters. The LCFG module generates the vertex index f_n for each building face conditioned on previously generated indices $f_{<n}$ with the vertex-centric local context $\mathcal{Z}^{(f)}$.

D. Model Training and Inference

During training, the negative log-likelihood was used as the loss function in the proposed method. In the GCVG module, the loss is calculated based on the predicted probabilities over vertex quantization categories and is defined as

$$\mathrm{V_Loss} = -\sum_{i=1}^{N_v} \log P\left(\mathcal{V}_t(i) \mid \mathcal{V}_{disp}\right) \tag{13}$$

where N_v denotes the number of valid vertices excluding padding. For each given ground-truth (GT) vertex $\mathcal{V}_t(i)$, the conditional log-likelihood is calculated under the predicted distribution $\mathcal{V}_{disp}$. In the LCFG module, the loss is calculated

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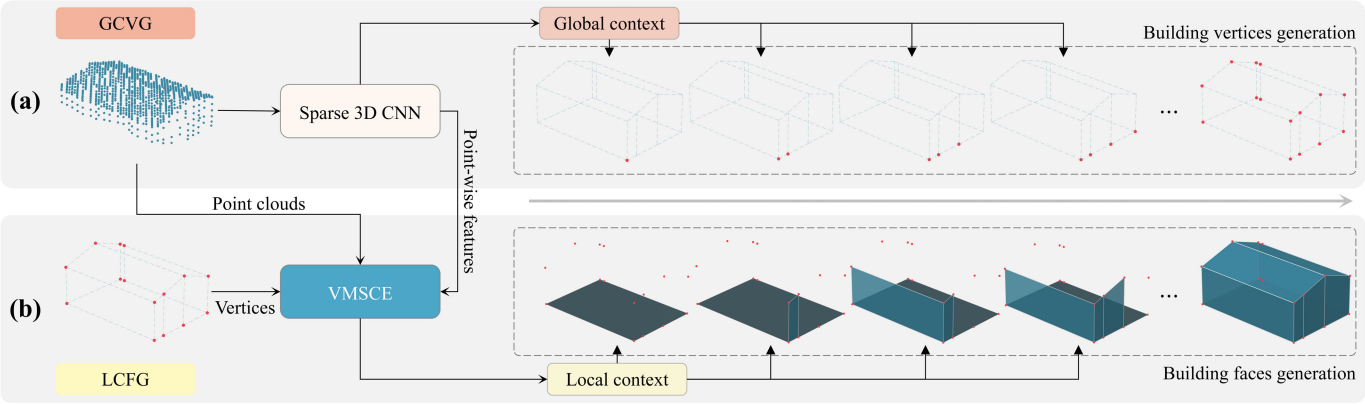


Fig. 5. Illustration of the building mesh generation process. (a) Building vertices generation conditioned on the global context. (b) Building face generation conditioned on the local context.



Fig. 6. Geographical distribution of (a) Zurich and (b) Tallinn datasets, where the test areas are highlighted with red boxes.

based on the predicted probabilities within the range of vertex indices, and is defined as

$$\text{F_Loss} = -\sum_{i=1}^{N_f} \log P\left(\mathcal{F}_t(i) \mid \mathcal{F}_{disp}\right) \tag{14}$$

where N_f denotes the number of valid vertex indices excluding padding. For each given GT vertex index $F_t(i)$, the conditional log-likelihood is calculated under the predicted distribution F_{disp}.

During inference, as illustrated in Fig. 5, the GCVG module encodes LiDAR PCs to obtain a global context, which conditions the generation of building vertices. Then, the VMSCE module uses the generated vertices as centers to query the PCs and their associated features, aggregating multiscale semantics to form local context. Finally, the LCFG module generates building faces conditioned by the local context, achieving building reconstruction. In addition, nucleus sampling [48] with a cumulative probability threshold of 0.9 was used to regulate the decoder output. The hard constraints and structural checks from P2B [22] were applied to prevent degradation of sample quality and improve the integrity and coherence of the generated building meshes.

IV. Experiments and Results

A. Dataset

Extensive experiments were conducted on two widely used datasets to evaluate the proposed method, including Zurich dataset [22] and Tallinn dataset [49]. As shown in Fig. 6, both datasets consist of detailed building meshes collected from urban areas in Europe. Zurich dataset includes 27 227 buildings represented as polygonal meshes, depicting dense city blocks with varied architectural styles and building heights. Tallinn dataset comprises 44 885 building samples represented

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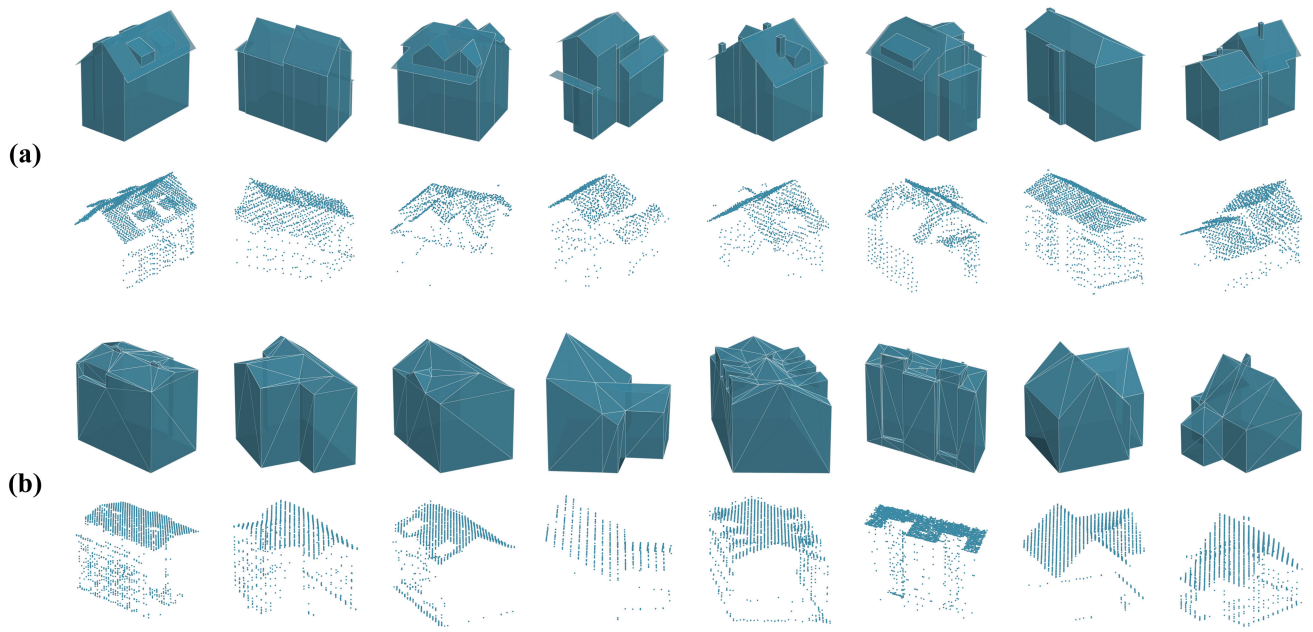


Fig. 7. Samples in (a) Zurich and (b) Tallinn datasets.

TABLE I
TRAINING CONFIGURATIONS OF GCVG AND LCFG MODULES, AND PARAMETER SETTINGS OF THE VMSCE MODULE

GCVG		LCFG		VMSCE	
Parameters	Value	Parameters	Value	Parameters	Value
Epoch	200	Epoch	200	Sampling layer depth	4
Batch size	16	Batch size	32	Dimension	384
Learning rate	3e-4	Learning rate	3e-4	Dropout rate	20%
Warm-up step	10000	Warm-up step	10000	Quantized bit	8
Dropout rate	20%	Dropout rate	20%		
Quantized bit	8	Quantized bit	8		
Dimension	384	Dimension	384		

by triangular meshes, covering both historical and modern districts with diverse spatial layouts and structural complexity. Fig. 7 presents several samples from Zurich and Tallinn datasets.

B. Implementation Details

During preprocessing, point coordinates were discretized using 8-bit quantization. The proposed method was implemented in PyTorch [50] and trained for 200 epochs. It was trained by the PyTorch lightning framework using the Adam optimizer [51], with a learning rate schedule comprising a warm-up phase followed by cosine annealing. For the network architecture, the embedding dimensions of all modules were set uniformly at 384. In addition, a node dropout rate of 20% was applied to improve generalization. The specific training configurations and other parameters are listed in Table I. Additional implementation details are available in our open-source repository.

The datasets used in the experiments went through a unified pipeline prior to training and inference. Samples with more than 100 building vertices were excluded, and PCs were uniformly downsampled to a maximum of 4096 points. Both the building meshes and the PCs were scaled, translated, and aligned to fit within a unit diagonal bounding box. To eliminate background noise, building mesh boundaries were slightly expanded, and surface PCs were retained based on Euclidean distance to 10 000 uniformly sampled mesh points. All methods are evaluated on the same filtered dataset, which is produced by consistently applying the filtering rules across all buildings and scenes, regardless of building complexity. During training, data augmentation was applied to improve generalization, including per-axis scaling (0.8–1.2), random rotation around the z-axis (0°–360°), and translation along each axis (−0.2 to 0.2). All augmentations were consistently applied to both mesh and PC data.

All experiments were conducted on an Intel Core i9-14900K CPU and an NVIDIA GeForce RTX 4090 GPU.

C. Evaluation Metrics

This study uses mesh surface point sampling to evaluate the performance of the proposed method, which is widely used to compare reconstructed and GT meshes [21], [52], [53]. The Chamfer distance (CD), Hausdorff distance (HD), and F1Score metrics are calculated between the sample points to evaluate the quality of the reconstruction results. In addition, two widely used metrics in generative method evaluation,

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i.e., minimum matching distance (MMD) and coverage (COV) [54], [55], [56], are also reported.

The CD quantifies the average nearest-neighbor distance between two sets of points, serving as a measure of geometric similarity. Two forms of CD are considered: $CD_{inp\to rec}$, which evaluates the distance from PCs to the reconstructed meshes, and $CD_{rec\to ref}$, which measures the distance from the reconstructed meshes to the GT meshes. The CD is defined as

$$\mathrm{CD}\ (\mathrm{X},\mathrm{Y}) = \frac{1}{|\mathrm{X}|}\sum_{x\in\mathrm{X}}\min_{y\in\mathrm{Y}}\ \|x-y\|^2 + \frac{1}{|\mathrm{Y}|}\sum_{y\in\mathrm{Y}}\min_{x\in\mathrm{X}}\ \|y-x\|^2 \tag{15}$$

where X and Y are two sets of nonempty points. A lower CD indicates a smaller difference between the two sets of points, reflecting a higher similarity between the reconstructed and GT meshes.

The HD quantifies the maximum distance from any point in one set to the nearest point in the other set, thereby measuring the greatest deviation between the reconstructed and GT meshes. The formula for calculating HD is as follows:

$$\mathrm{HD}\ (\mathrm{X},\mathrm{Y}) = \max\left\{\sup_{x\in\mathrm{X}}\left(\inf_{y\in\mathrm{Y}} d\,(x,y)\right), \sup_{y\in\mathrm{Y}}\left(\inf_{x\in\mathrm{X}} d\,(x,y)\right)\right\} \tag{16}$$

where sup refers to the least upper bound (supremum), which is the smallest value greater than or equal to every element, while inf denotes the greatest lower bound (infimum), the largest value less than or equal to every element, both measured in terms of Euclidean distance $d(\cdot)$.

Furthermore, F1Score is calculated between the two sets of points at an error distance threshold τ to evaluate the reconstruction performance. The F1Score is defined as

$$\mathrm{F1Score}\,(\tau) = \frac{2\cdot \mathrm{P}\,(\tau)\cdot \mathrm{R}\,(\tau)}{\mathrm{P}\,(\tau)+\mathrm{R}\,(\tau)} \tag{17}$$

where $\mathrm{P}(\tau)$ and $\mathrm{R}(\tau)$ represent the precision and recall, respectively, under the error distance threshold τ

$$\mathrm{P}\,(\tau) = \frac{1}{|\mathrm{S}_1|}\sum_{r\in\mathrm{S}_1}\left[\min_{t\in\mathrm{S}_2}\|t-r\| < \tau\right] \tag{18}$$

$$\mathrm{R}\,(\tau) = \frac{1}{|\mathrm{S}_2|}\sum_{r\in\mathrm{S}_2}\left[\min_{t\in\mathrm{S}_1}\|r-t\| < \tau\right] \tag{19}$$

where S_1 represents the set of reconstructed points, S_2 is the set of GT points, and $\tau = 0.05$ is the tolerable error distance threshold. A sampled point on the reconstructed building mesh surface is selected when its distance to the corresponding point on the GT is less than τ.

D. Baselines

Five representative building reconstruction methods were selected as baselines to evaluate the performance of the proposed method, including 2.5-D dual contour (2.5DC) [57], City3D [58], P2B [22], PolyGNN [59], and BWFormer [39]. Specifically, 2.5DC is a classic dual contouring-based method that optimizes both 3-D geometric surfaces and 2-D roof boundaries to reconstruct building models with arbitrary roof shapes. City3D reconstructs roof structures and vertical walls from PCs by performing plane extraction and topology optimization. PolyGNN employs graph neural networks to learn the assembly of primitives extracted from PCs, facilitating the reconstruction of watertight 3-D building models. BWFormer designs a 2-D-to-3-D corner detection strategy to improve retrieval efficiency and adopts a discriminative approach for building reconstruction. P2B is a generative method that reconstructs buildings by first generating vertices from PCs, followed by face generation based on the generated vertices. Its token-level constraint strategy enhances the structural integrity of the results.

E. Overall Results

Table II reports the quantitative evaluation results of all comparative methods on Zurich and Tallinn datasets. On Zurich dataset, the proposed method achieved the lowest $CD_{rec\to ref}$ of 6.75 cm and the highest F1Score of 85.94%, outperforming all the baseline methods. These results demonstrate the superior accuracy and completeness of the reconstructed meshes. In addition, the proposed method achieved the lowest HD of 10.70 cm, the second-highest COV of 67.62%, and the lowest MMD of 2.37 mm, reflecting both improved accuracy and surface smoothness. Although P2B obtained slightly better COV, the overall balance of the proposed method across all metrics demonstrates superior performance. A similar trend is observed on Tallinn dataset. The proposed method achieved the best $CD_{rec\to ref}$ of 5.79 cm, improving upon the best baseline City3D by 0.51 cm. It also achieves the highest F1Score of 91.15%, surpassing all baseline methods. Moreover, the proposed method yields competitive results across other metrics, including HD of 11.46 cm, the highest COV, and the lowest MMD. These quantitative results suggest that the proposed method improves the accuracy of building reconstruction by integrating the spatial context inherent in PCs.

To qualitatively evaluate the performance of the proposed method, Fig. 8 visualizes the reconstruction results of several representative samples from Zurich and Tallinn datasets. PolyGNN is not shown in the figure because the method failed to successfully reconstruct any of the selected samples. In the figure, the last row of each subfigure shows the GT building meshes for reference, while the first row displays the input PCs of the samples. The samples are typically sparse and contain limited geometric information, which poses significant challenges for accurate 3-D reconstruction.

Fig. 8 shows that the proposed method on the two datasets demonstrates superior performance in terms of structural completeness, geometric accuracy, and overall shape fidelity. The reconstructed meshes of the proposed method closely resemble the GT building meshes, which effectively capture complex architectural features such as chimneys, dormers, and protruding wall structures. In comparison, the 2.5DC method tends to produce overly fragmented surfaces, while the City3D method oversimplifies building meshes and fails to retain critical local details. BWFormer produces incomplete reconstructions, which can be attributed to the limited precision of its discriminative wireframe predictions. Although P2B adopts a generative paradigm, it still suffers from mesh holes, distortions, and missing components when dealing with

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TABLE II

EXPERIMENTAL RESULTS OF THE PROPOSED AND BASELINE METHODS ON ZURICH AND TALLINN DATASETS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD, WHILE THE SECOND-BEST RESULTS ARE UNDERLINED

Dataset	Method	$CD_{inp\to rec}$ (cm)↓	$CD_{rec\to ref}$ (cm)↓	HD (cm)↓	F1Score (%)↑	COV (%)↑	MMD (mm)↓
Zurich	2.5DC	20.19	10.74	24.95	62.83	40.59	5.04
	City3D	21.75	8.71	12.49	74.80	49.13	2.96
	PolyGNN	24.99	10.53	15.57	69.73	54.27	3.42
	BWFormer	21.73	11.78	26.23	57.56	45.37	6.22
	P2B	19.83	7.44	13.76	82.97	**69.77**	2.48
	Ours	**19.65**	**6.75**	**10.70**	**85.94**	67.62	**2.37**
Tallinn	2.5DC	**11.10**	9.65	26.56	70.70	56.82	5.85
	City3D	11.62	6.30	**11.23**	88.74	78.68	2.26
	PolyGNN	18.17	14.04	21.61	65.14	62.77	3.40
	BWFormer	17.16	12.46	28.12	52.97	40.87	7.09
	P2B	13.43	7.67	17.55	79.85	68.88	3.11
	Ours	12.06	**5.79**	11.46	**91.15**	**86.38**	**2.10**

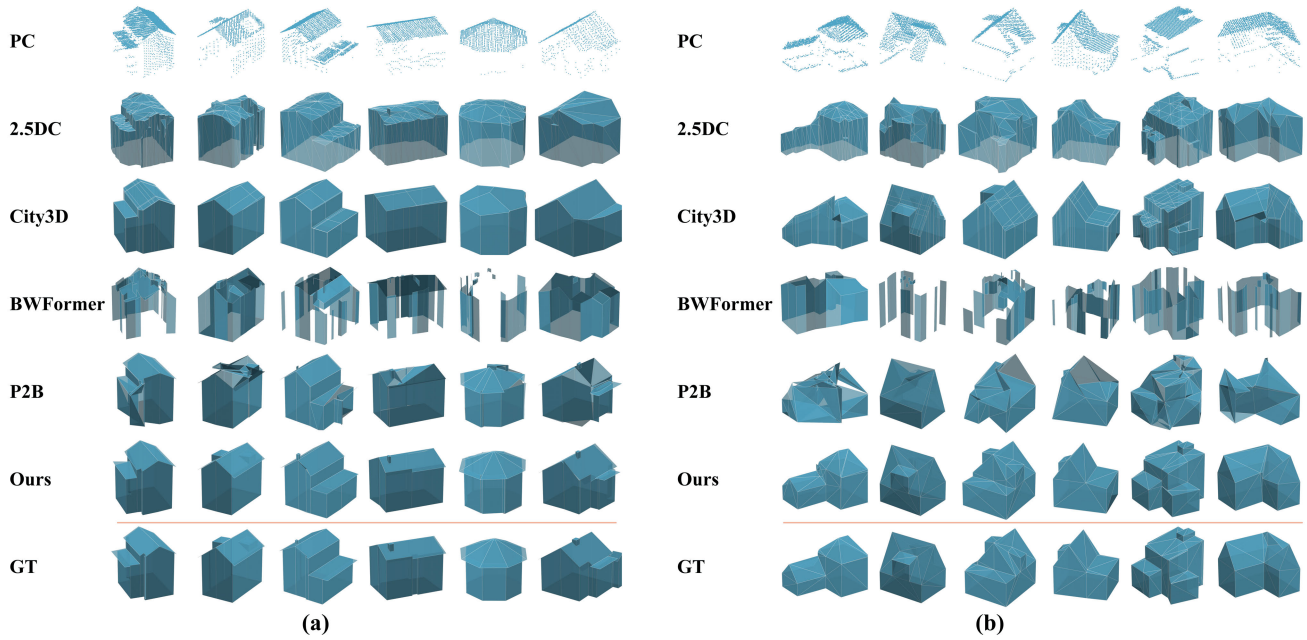


Fig. 8. Samples of the reconstruction results on (a) Zurich and (b) Tallinn datasets.

TABLE III

ABLATION STUDY RESULTS ON ZURICH DATASET

Method	$CD_{rec\to ref}$ (cm)↓	HD (cm)↓	F1Score (%)↑
Full	**6.75**	**10.70**	**85.94**
w/o GCVG	14.76	20.55	29.21
w/o LCFG	13.44	29.31	56.62
w/o VMSCE	6.94	11.13	85.07

TABLE IV

EXPERIMENTAL RESULTS OF SAMPLING LAYER DEPTH IN VMSCE MODULE ON ZURICH DATASET

#Layer	$CD_{rec\to ref}$ (cm)↓	HD (cm)↓	F1Score (%)↑
1	6.848	10.920	85.222
2	6.774	10.748	85.750
3	6.784	10.850	85.609
4	**6.754**	**10.698**	**85.939**
5	6.791	10.840	85.608
6	6.786	10.865	85.623

complex building structures. In general, the proposed method exhibits excellent geometric fidelity and structural integrity, which further validates the effectiveness of integrating the spatial context with the generative model.

V. DISCUSSION

A. Ablation Study

To evaluate the contribution of three key modules, ablation experiments on Zurich dataset were conducted by systematically removing each of them from the proposed method. Specifically, a discriminative method [39] was adopted when either the GCVG or LCFG module was removed, and was used for vertex extraction and face prediction, respectively.

As shown in Table III, the proposed full model achieves the best overall performance, with a $CD_{rec\to ref}$ of 6.75 cm, an HD of 10.70 cm, and an F1Score of 85.94%. Removing the GCVG module leads to performance degradation across all metrics, where $CD_{rec\to ref}$ and HD increase to 14.76 and

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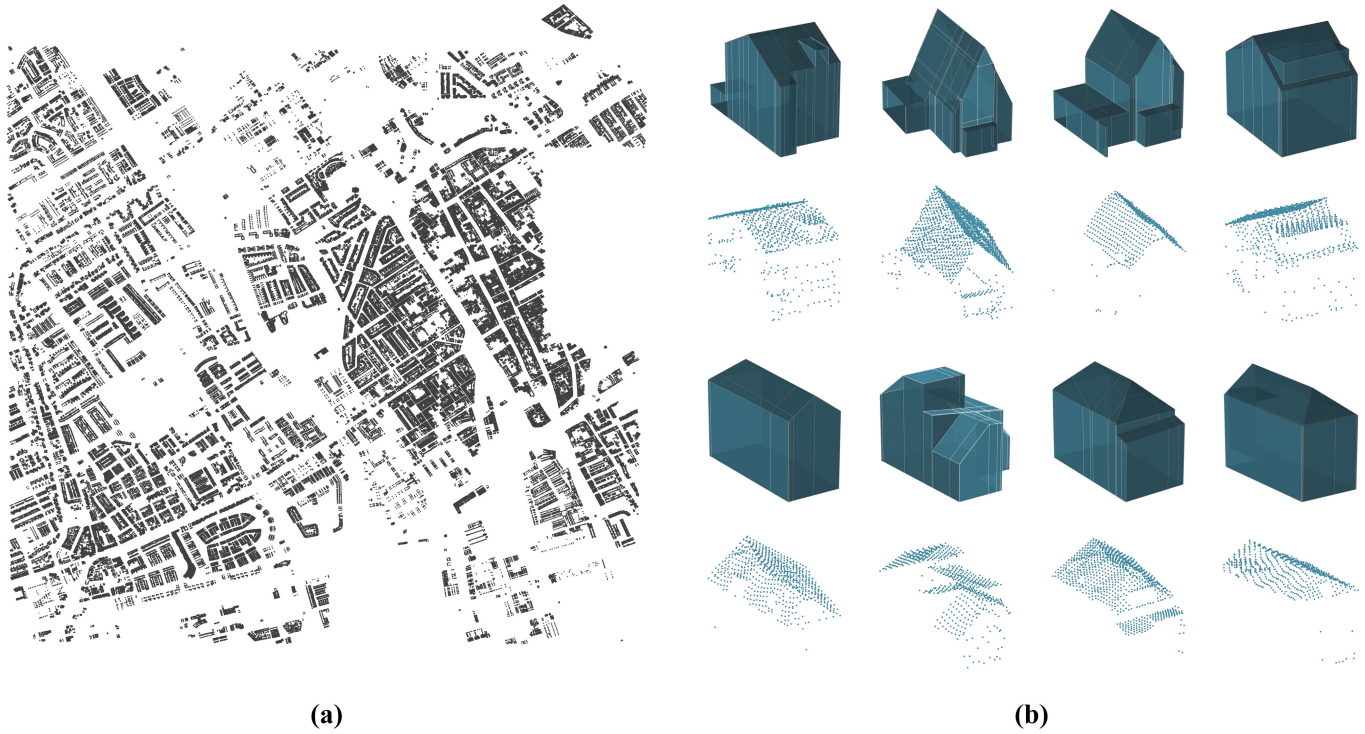


Fig. 9. Illustration of AHN3: (a) dataset region and (b) samples.

TABLE V
CROSS-SCENE EVALUATION RESULTS ON THE AHN3 DATASET. MODELS ARE TRAINED ON ZURICH OR TALLINN AND TESTED ON AHN3

Setting	Method	$CD_{inp\to rec}$ (cm)↓	$CD_{rec\to ref}$ (cm)↓	HD (cm)↓	F1Score (%)↑
Zurich→AHN3	P2B	19.06	8.12	15.56	77.38
	Ours	**18.67**	**7.28**	**12.66**	**81.70**
Tallinn→AHN3	P2B	22.17	10.50	22.38	67.57
	Ours	**19.92**	**7.77**	**13.84**	**79.57**

20.55 cm, respectively, and F1Score drops to 29.21%. This is because, in the absence of global-context constraints, the overall geometric deviations are amplified, leading to severe degradation of structural integrity. This demonstrates that the GCVG module plays a decisive role in keeping the structural integrity of building models. Removing the LCFG module also results in performance degradation, particularly in terms of HD and F1Score. The results indicate that the generative face construction mechanism of the LCFG module is crucial for improving geometric reconstruction quality. After removing the VMSCE module, $CD_{rec\to ref}$, HD, and F1Score all exhibit consistent degradation, suggesting that without multiscale vertex-centric local context, the geometric semantics during face generation are weakened. This demonstrates that the VMSCE module improves the accuracy of face generation by integrating a richer spatial context. Overall, the GCVG, LCFG, and VMSCE modules work collaboratively to enhance building reconstruction.

B. Parameter Sensitivity Analysis

In the VMSCE module, the depth of sampling layers is the critical hyperparameter for multiscale encoding. A sensitivity analysis of this parameter was conducted on Zurich dataset by varying the sampling depth, with quantitative results reported in Table IV. The proposed method achieves optimal performance when the depth is set to four, reaching an F1Score of 85.939%, HD of 10.698 cm, and $CD_{rec\to ref}$ of 6.754 cm. It can be explained by the fact that when there are few layers, it does not involve enough points to fully capture the semantics of spatial context centered on building vertices; conversely, when the depth of layers is set to larger than four, it involves more points, introducing excessive noise.

C. Experiment on Cross-Scene Reconstruction

To evaluate the generalization capability of the proposed method, two cross-scene reconstruction experiments were conducted, with P2B used as the comparison method. The model was trained on Zurich and Tallinn datasets, respectively, and evaluated on the AHN3 dataset [58], which contains approximately 20 000 building pairs, as illustrated in Fig. 9. As reported in Table V, under Zurich→AHN3 setting, the proposed method outperforms P2B by achieving lower $CD_{rec\to ref}$ and HD values while increasing the F1Score from 77.38% to 81.70%. In the more challenging Tallinn→AHN3 setting,

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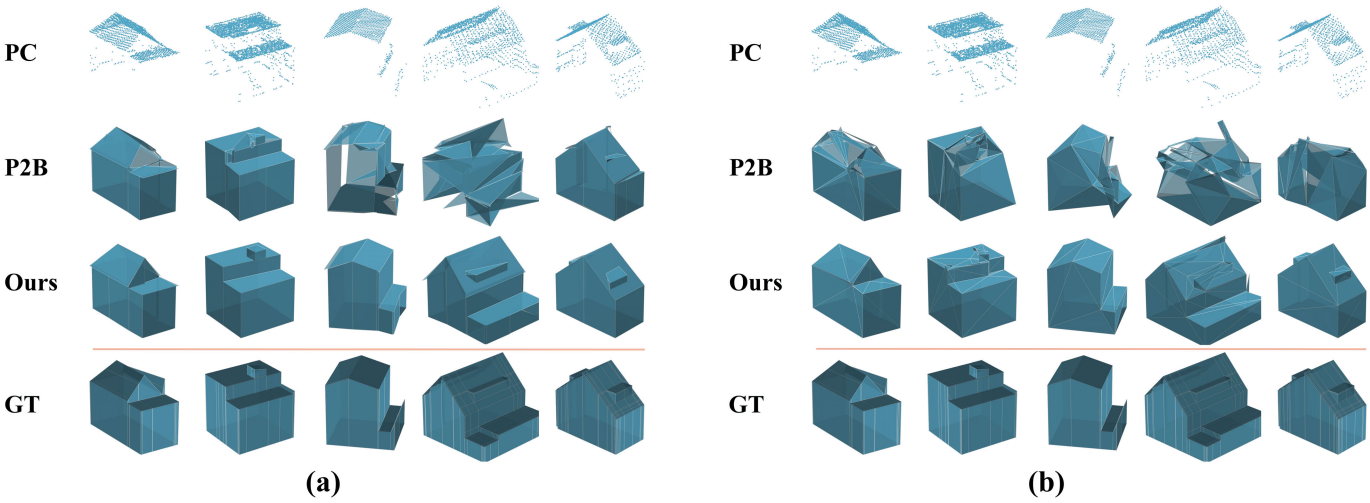


Fig. 10. Samples of reconstruction results on the AHN3 dataset, where the model is trained on (a) Zurich or (b) Tallinn dataset.

TABLE VI
EXPERIMENTAL RESULTS OF METHOD EFFICIENCY ON ZURICH DATASET

Method	P (M)	|V|	|F|	T (s)↓
2.5DC	-	84.52	133.97	0.02
City3D	-	158.49	56.84	0.77
PolyGNN	3.48	31.48	27.58	0.37
BWformer	62.09	26.94	14.41	0.11
P2B	80.47	37.43	19.95	1.80
Ours	121.80	33.97	18.29	1.31

similar improvements are observed, with substantial reductions in $CD_{rec\rightarrow ref}$ and HD and a notable F1Score increase from 67.57% to 79.57%. These results demonstrate strong cross-scene generalization capability and robust geometric reconstruction quality across different urban scenarios.

Fig. 10 presents several reconstruction samples on the AHN3 dataset using models trained on Zurich or Tallinn dataset, respectively, where the results of the proposed method are shown in the third row. As shown in the second row, while the P2B method recovers the overall building shapes at the vertex generation level, its face generation is often flawed, leading to incomplete and lower quality reconstruction results. In contrast, the proposed method consistently generates building faces that closely resemble the GT, even across datasets with different mesh representations, such as polygonal faces in Zurich and triangular meshes in Tallinn. It can be attributed to the integration of the global–local context, which balances overall structural constraints and local geometric details, enabling the model to adapt across diverse scenarios.

D. Method Efficiency

The efficiency of the proposed method and the baselines is evaluated based on the number of mesh vertices and faces (|V| and |F|), the inference time (T), and model parameters (P). The results on Zurich dataset are shown in Table VI, and the proposed method generates concise building meshes with fewer vertices and faces compared to most baselines.

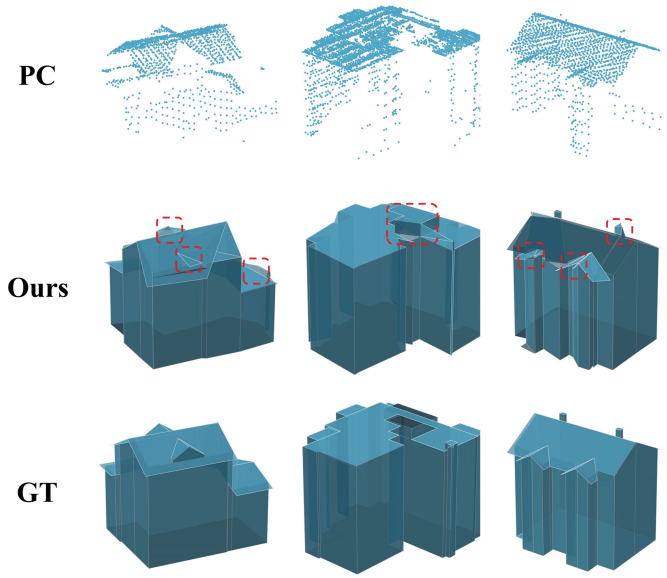


Fig. 11. Examples of failure cases with geometric reconstruction errors.

In particular, although the proposed method contains more learnable parameters than P2B, it yields a shorter inference time of 1.31 s. This advantage is attributed to improved reconstruction accuracy, which reduces the number of iterations required for structural checks. Overall, the results demonstrate that the proposed method improves the accuracy of building reconstruction while maintaining great computational efficiency.

E. Failure Cases

To further investigate the limitations of the proposed method, several failure cases are visualized in Fig. 11. As highlighted by the red boxes in the figure, face misalignment errors occur in regions with intricate geometric details. This can be attributed to the diverse PC distribution coupled with insufficient training samples in complex regions, resulting in poor adaptability of the model to intricate geometric scenes.

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VI. Conclusion

This article presents a novel method named SCGNet for 3-D building reconstruction from airborne LiDAR PCs by integrating spatial context with the generative model. The proposed method extracts vertex-centric context from PCs and integrates a Transformer-based network with global–local context modeling to improve the accuracy of 3-D building reconstruction. Extensive experiments demonstrate that the proposed method considerably improves building reconstruction performance and exhibits strong generalization in cross-scene reconstruction tasks. This study explores a novel approach to optimizing building reconstruction and serves as a methodological reference for PC-based 3-D model tasks.

Future work could explore more efficient geometry-aware modeling strategies to further improve generalization across diverse PC distributions. In addition, new approaches are expected to address the challenges of geometrically complex regions by more comprehensively capturing local features and expanding the diversity of training data, thereby enhancing overall reconstruction fidelity.

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